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1 "/Users/son/Library/Mobile Documents/com~apple~CloudDocs/University/Laboratory/Year 3 Lab/pycharm_venv/bin/python3" /Users/son/Library/Mobile Documents/com~apple~CloudDocs/University/Laboratory/Year 3 Lab/Data_Analysis/main.py
2 Enter the path to the video (with A0): /Users/son/Library/Mobile Documents/com~apple~CloudDocs/University/Laboratory/Year 3 Lab/Data_Analysis/Videos/with_A0/3ohm/0.9A_3R_with_A0.avi
3 Processing Video
4 Title: 0.9A_3R_with_A0, Current: 0.9, Resistance: 3, With A0: True
5 Enter the stage of the process: 1
6 Initial 300 frames extracted
7 Amplitude: 52897.484556708376, xo: 803.5197859071487, yo: 718.4008514682954, sigma_x: 2.2417764431266813, sigma_y: 1.949767907616556, theta: 1.4693953924571452, offset: 0.33224489967220394
8 Function: 52897.484556708376 * exp(-((cos(theta)^2)/(2*sigma_x^2) + (sin(theta)^2)/(2*sigma_y^2))*(x-xo)^2 + 2*(-sin(2*theta)/(4*sigma_x^2) + sin(2*theta)/(4*sigma_y^2))*(x-xo)*(y-yo) + ((sin(theta)^2)/(2*sigma_x^2) + (cos(theta)^2)/(2*sigma_y^2))*(y-yo)^2) + offset
9 ROI defined
10 2025-01-21 09:59:23.726 python3[66358:53963084] +[IMKClient subclass]: chose IMKClient_Modern
11 2025-01-21 09:59:23.726 python3[66358:53963084] +[IMKInputSession subclass]: chose IMKInputSession_Modern
12 Processed 100 frames
13 SNR: 68.16898345947266
14 Processed 200 frames
15 SNR: 79.16909790039062
16 Processed 300 frames
17 SNR: 88.42523193359375
18 Processed 400 frames
19 SNR: 98.87196350097656
20 Processed 500 frames
21 SNR: 129.5682373046875
22 Processed 600 frames
```

23 SNR: 89.94303894042969
24 Processed 700 frames
25 SNR: 111.97354125976562
26 Processed 800 frames
27 SNR: 90.20635986328125
28 Processed 900 frames
29 SNR: 125.42048645019531
30 Processed 1000 frames
31 SNR: 51.48456573486328
32 Processed 1100 frames
33 SNR: 36.33560562133789
34 Processed 1200 frames
35 SNR: 33.231651306152344
36 Processed 1300 frames
37 SNR: 59.068912506103516
38 Processed 1400 frames
39 SNR: 52.94350051879883
40 Processed 1500 frames
41 SNR: 65.83021545410156
42 Processed 1600 frames
43 SNR: 84.2395248413086
44 Processed 1700 frames
45 SNR: 79.99999237060547
46 Processed 1800 frames
47 SNR: 67.47805786132812
48 Processed 1900 frames
49 SNR: 108.09222412109375
50 Processed 2000 frames
51 SNR: 71.76851654052734
52 Processed 2100 frames
53 SNR: 67.67538452148438
54 Processed 2200 frames
55 SNR: 76.49748229980469
56 Processed 2300 frames
57 SNR: 56.85258483886719
58 Processed 2400 frames
59 SNR: 52.48442840576172
60 Processed 2500 frames
61 SNR: 70.7761001586914
62 Processed 2600 frames
63 SNR: 29.340621948242188

64 Processed 2700 frames
65 SNR: 56.16898727416992
66 Processed 2800 frames
67 SNR: 84.98817443847656
68 Processed 2900 frames
69 SNR: 28.63606071472168
70 Processed 3000 frames
71 SNR: 75.05056762695312
72 Processed 3100 frames
73 SNR: 77.62637329101562
74 Processed 3200 frames
75 SNR: 82.50605010986328
76 Processed 3300 frames
77 SNR: 99.89369201660156
78 Processed 3400 frames
79 SNR: 91.23799133300781
80 Processed 3500 frames
81 SNR: 87.70082092285156
82 Processed 3600 frames
83 SNR: 101.01787567138672
84 Processed 3700 frames
85 SNR: 58.877593994140625
86 Processed 3800 frames
87 SNR: 76.12669372558594
88 Processed 3900 frames
89 SNR: 83.48986053466797
90 Processed 4000 frames
91 SNR: 82.43258666992188
92 Processed 4100 frames
93 SNR: 54.89654541015625
94 Processed 4200 frames
95 SNR: 69.00347137451172
96 Processed 4300 frames
97 SNR: 69.98467254638672
98 Processed 4400 frames
99 SNR: 42.81237030029297
100 Processed 4500 frames
101 SNR: 54.5576057434082
102 Processed 4600 frames
103 SNR: 60.74212646484375
104 Processed 4700 frames

105 SNR: 69.20301818847656
106 Processed 4800 frames
107 SNR: 24.126745223999023
108 Processed 4900 frames
109 SNR: 13.776482582092285
110 Processed 5000 frames
111 SNR: 37.995059967041016
112 Processed 5100 frames
113 SNR: 44.71337127685547
114 Processed 5200 frames
115 SNR: 24.505264282226562
116 Processed 5300 frames
117 SNR: 81.14138793945312
118 Processed 5400 frames
119 SNR: 58.31159210205078
120 Processed 5500 frames
121 SNR: 58.636680603027344
122 Processed 5600 frames
123 SNR: 33.540584564208984
124 Processed 5700 frames
125 SNR: 22.157737731933594
126 Processed 5800 frames
127 SNR: 84.36685943603516
128 Processed 5900 frames
129 SNR: 42.437374114990234
130 Processed 6000 frames
131 SNR: 80.63697814941406
132 Processed 6100 frames
133 SNR: 92.06631469726562
134 Processed 6200 frames
135 SNR: 55.554115295410156
136 Processed 6300 frames
137 SNR: 41.9962043762207
138 Processed 6400 frames
139 SNR: 48.866180419921875
140 Processed 6500 frames
141 SNR: 42.42102813720703
142 Processed 6600 frames
143 SNR: 68.89366149902344
144 Processed 6700 frames
145 SNR: 116.254638671875

146 Processed 6800 frames
147 SNR: 66.92627716064453
148 Processed 6900 frames
149 SNR: 56.74879455566406
150 Processed 7000 frames
151 SNR: 82.59300994873047
152 Processed 7100 frames
153 SNR: 99.49466705322266
154 Processed 7200 frames
155 SNR: 74.70658111572266
156 Processed 7300 frames
157 SNR: 29.902385711669922
158 Processed 7400 frames
159 SNR: 66.5655517578125
160 Processed 7500 frames
161 SNR: 24.104167938232422
162 Processed 7600 frames
163 SNR: 10.819164276123047
164 Processed 7700 frames
165 SNR: 37.962158203125
166 Processed 7800 frames
167 SNR: 14.694343566894531
168 Processed 7900 frames
169 SNR: 26.05317497253418
170 Processed 8000 frames
171 SNR: 59.792930603027344
172 Processed 8100 frames
173 SNR: 31.51238441467285
174 Processed 8200 frames
175 SNR: 67.71595001220703
176 Processed 8300 frames
177 SNR: 51.277976989746094
178 Processed 8400 frames
179 SNR: 54.88508987426758
180 Processed 8500 frames
181 SNR: 42.333194732666016
182 Processed 8600 frames
183 SNR: 42.9936637878418
184 Processed 8700 frames
185 SNR: 20.867277145385742
186 Processed 8800 frames

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187 SNR: 25.700515747070312
188 Processed 8900 frames
189 SNR: 48.891605377197266
190 Processed 9000 frames
191 SNR: 37.254695892333984
192 Processed 9100 frames
193 SNR: 30.127498626708984
194 Processed 9200 frames
195 SNR: 34.41566467285156
196 Processed 9300 frames
197 SNR: 23.12305450439453
198 Processed 9400 frames
199 SNR: 26.867633819580078
200 Processed 9500 frames
201 SNR: 18.933536529541016
202 Processed 9600 frames
203 SNR: 18.581327438354492
204 Processed 9700 frames
205 SNR: 22.690032958984375
206 Processed 9800 frames
207 SNR: 35.551246643066406
208 Processed 9900 frames
209 SNR: 24.286718368530273
210 Processed 10000 frames
211 SNR: 33.1331787109375
212 Processed 10100 frames
213 SNR: 34.36516189575195
214 Processed 10200 frames
215 SNR: 28.43141746520996
216 Processed 10300 frames
217 SNR: 64.67472076416016
218 Processed 10400 frames
219 SNR: 61.03791427612305
220 Processed 10500 frames
221 SNR: 39.256080627441406
222 Processed 10600 frames
223 SNR: 28.783933639526367
224 Processed 10700 frames
225 SNR: 23.291610717773438
226 Saving Signals and Signal_errs
227 Saving Signal_errs
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228 Saving Time
229 Saving SNRs
230 Saving SNR_errs
231 Rolling AVG = 11
232
233 Process finished with exit code 0
234
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